ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 6

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE A

MARKS: 70 **TIME: 1½ hours**

An end-of-year test on Grade 6 Terms 3 and 4: Energy and Change (electric circuits, conductors and insulators, mains electricity) and Planet Earth and Beyond (the solar system, movements of the Earth and the Moon, telescopes, exploring the Moon and Mars).

Written by Academic Central for practice, following the CAPS curriculum for Grade 6. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully before you answer.
4. Use a pencil and a ruler for the circuit diagram in QUESTION 3.
5. Diagrams are NOT drawn to scale.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which of these must every working circuit have?

A A switch
B A source of energy, such as a cell
C A buzzer
D Two bulbs

1.2 Which material is a good conductor of electricity?

A Plastic B Wood C Copper D Rubber

1.3 The switch in a circuit is used to …

A make the bulb brighter.
B open and close the circuit.
C store energy.
D change the colour of the light.

1.4 Most of South Africa's electricity is made in power stations that burn …

A coal. B wood. C paper. D paraffin.

1.5 Electricity travels from power stations to our homes through …

A water pipes.
B pylons and power lines.
C rivers.
D gas pipes.

1.6 The planet closest to the Sun is …

A Venus. B Earth. C Mercury. D Mars.

1.7 The Earth spinning on its own axis is called …

A revolution. B rotation. C orbit. D eclipse.

1.8 When we cannot see the Moon at all in the night sky, it is called a …

A full moon. B new moon. C half moon. D crescent moon.

1.9 An instrument that makes faraway objects in space look bigger and closer is a …

A microscope. B telescope. C periscope. D stethoscope.

1.10 The "red planet" that robot rovers are exploring is …

A Jupiter. B Saturn. C Mars. D Neptune.

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QUESTION 2: SCIENTIFIC WORDS

Write the correct scientific word for each description. Write only the word next to the question number.

2.1 A material that does not allow electricity to flow through it

2.2 The complete path along which electricity flows

2.3 Two or more cells connected together

2.4 Joining to mains electricity without permission, which is dangerous and against the law

2.5 The path that a planet follows around the Sun

2.6 The time it takes the Earth to move once around the Sun

2.7 A natural object, like our Moon, that moves around a planet

2.8 A person who is trained to travel into space

[8]

QUESTION 3: ELECTRIC CIRCUITS

3.1 Study the circuit diagram in FIGURE 1.

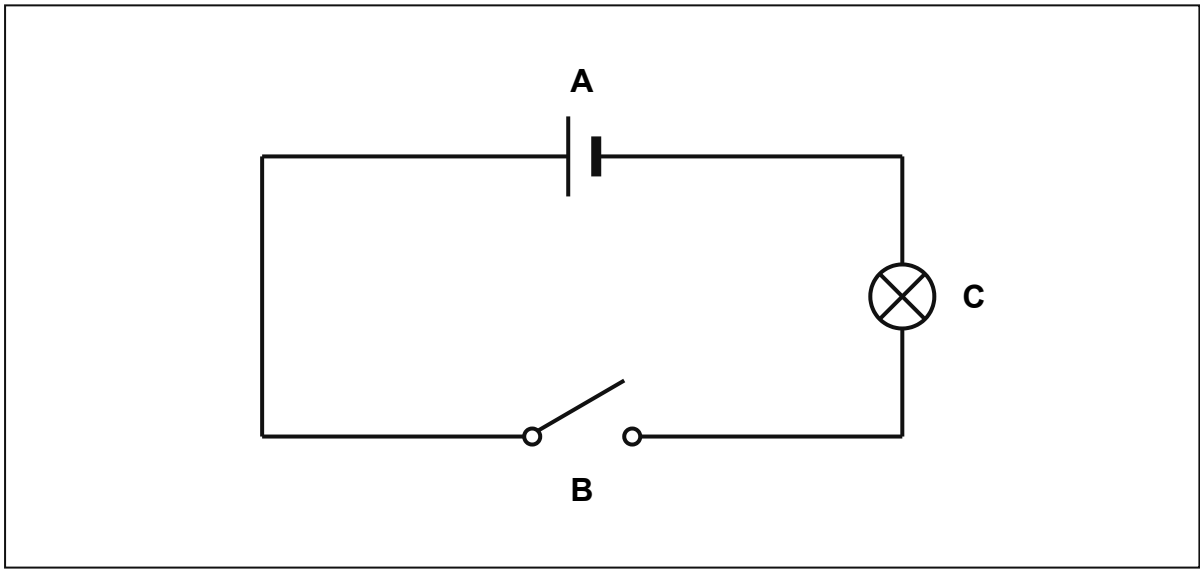


FIGURE 1: A circuit diagram

3.1.1 Name the parts labelled A, B and C. (3)

3.1.2 Will the bulb light up in this circuit? Give a reason for your answer. (2)

3.1.3 What is the job of part A in the circuit? (1)

3.2 Draw a circuit diagram, using the correct symbols, of a circuit with TWO cells, a CLOSED switch and ONE bulb. (4)

3.3 Sizwe tested these objects in a circuit to see which ones would make the bulb light up: a steel nail, a plastic ruler, a piece of aluminium foil, a rubber eraser, a wooden spoon and a coin.

3.3.1 Which THREE objects made the bulb light up? (3)

3.3.2 What do we call materials that allow electricity to flow through them? (1)

3.3.3 Why are the copper wires in electric cables covered with plastic? (2)

3.4 Give ONE way to make the bulb in a circuit shine more brightly. (1)

3.5 A learner closes the switch in a circuit, but the bulb does not light up. Give ONE possible reason. (1)

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QUESTION 4: MAINS ELECTRICITY

4.1 The steps below show how electricity is made in a coal power station, but they are mixed up. Write the letters in the correct order. (3)

A The steam turns the turbines.
B Coal is burnt to heat water.
C The turning turbines drive a generator, which makes electricity.
D The water boils and turns into steam.
E The electricity travels along power lines to towns and cities.

4.2 Give TWO ways that burning coal to make electricity harms the environment. (2)

4.3 Name TWO renewable sources that can be used to make electricity. (2)

4.4 Give TWO reasons why illegal connections are dangerous. (2)

4.5 Give THREE ways your family can save electricity at home. (3)

[12]

QUESTION 5: PLANET EARTH AND BEYOND

5.1 FIGURE 2 shows the Sun and the eight planets, numbered 1 to 8.

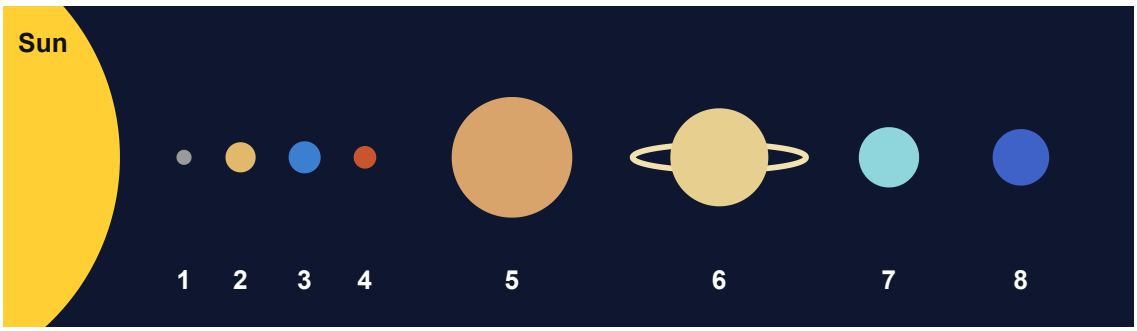


FIGURE 2: The solar system (not to scale)

5.1.1 Name planet number 3. (1)

5.1.2 Which planet is the biggest? Give its name and its number. (2)

5.1.3 Name planet number 6, the planet with large rings. (1)

5.1.4 Why are planets 7 and 8 very cold? (1)

5.1.5 Name the FOUR rocky planets. (2)

5.2 Explain the difference between the *rotation* and the *revolution* of the Earth. Say how long each one takes. (4)

5.3 FIGURE 3 shows four phases of the Moon, A to D.

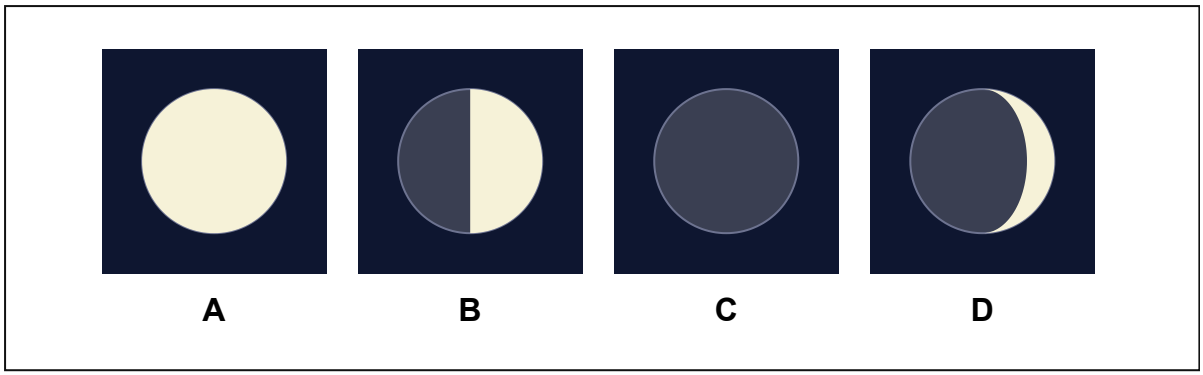


FIGURE 3: Four phases of the Moon

5.3.1 Name phase A and phase C. (2)

5.3.2 Explain why the Moon seems to change its shape during the month. (2)

5.3.3 About how long does the Moon take to move once around the Earth? (1)

5.4 Give ONE thing that scientists have learnt about space by using telescopes. (1)

5.5 Give TWO reasons why people cannot live on the Moon or Mars without special equipment. (2)

5.6 Why do scientists send robots (rovers) to Mars instead of people? (1)

5.7 Name TWO things that astronauts must take with them into space to stay alive. (2)

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TOTAL: 70

END OF QUESTION PAPER